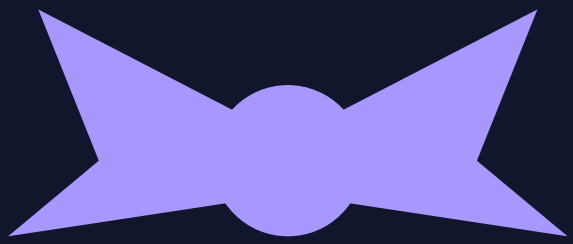
WORLD OF BATS

Python lab guide | Linux | 2026

DESIGN | CODE | HUNT

Build your own artificially intelligent bat, teach it to use echolocation, and compete to become the most efficient insect hunter.

University of Bristol | School of Computer Science



Your mission

Create a bat that can find and eat flying insects. You will customise how it looks, then improve its hunting strategy by changing Python code. You can work alone or with a partner.

TODAY	There is no single perfect answer. Try an idea, run it, watch what happens, then improve it.

Before you start

1	Use the Linux computer you were given and open this guide in the browser.
2	Download **World-of-Bats-2026-Linux.zip** from the workshop website.
3	Ask a demonstrator whenever you are unsure. You do not need previous Python experience.

What you will make

LOOK	THINK	COMPETE
Choose wing shape, body shape, size and colour.	Use echolocation data to decide when to turn.	Run bats together and compare their insect scores.

KEYS	Press **Q** to close the bat world. Press **Tab** to show or hide each bat's score.

1. Download and set up

1

Open a Terminal

Open the applications menu, search for **Terminal**, then open it. A Terminal lets you give short text commands to the computer.

2

Go to Downloads

Type the command below and press Enter.

```
cd ~/Downloads
```

3

Unzip the lab

Type both commands. The first extracts the download; the second enters the new folder.

```
unzip -q World-of-Bats-2026-Linux.zip
cd World-of-Bats-2026
```

4

Install the two Python libraries

These commands make the setup file runnable, then run it. Installation can take a minute.

```
chmod +x run-on-linux.sh
./run-on-linux.sh
```

5

Start the bat world

Run the main Python file.

```
python3 WorldOfBats.py
```

SUCCESS	A new window opens with bats flying over a forest. Press **Q** to close it before editing your code.

NEED HELP?	If you see red text, do not worry. Leave it on screen and show a demonstrator.

2. Make the bat yours

1

Rename your bat folder

Open **World-of-Bats-2026/BatCave**. Rename **YourBatsName** using your first name or a team name. Use letters and numbers only - no spaces.

2

Open only MyBat.py

Inside your renamed folder, open **MyBat.py** in the text editor. Keep this filename exactly as it is.

3

Choose wings

The first four bits control size (0000 smallest to 1111 largest). The final four bits choose the style.

```
self.setWingAppearance("11110100")
```

0000 	0001 	0010 
0011 	0100 	0101 

4

Choose a body

Use the same eight-bit pattern: four size bits followed by four style bits.

```
self.setBodyAppearance("10000011")
```

0000 	0001 	0010 
0011 	0100 	0101 

3. Add colour and test

Colour is written as three numbers: red, green and blue. Each number must be between 0 and 255.

Red	Green	Blue	Grey	Orange
(255, 0, 0)	(0, 255, 0)	(0, 0, 255)	(120, 120, 120)	(236, 115, 57)

Put your chosen values into MyBat.py:

```
self.setWingAppearance("11110100")
self.setBodyAppearance("10000011")
self.setColour((236, 115, 57))
```

1

Save

Press **Ctrl+S** in the text editor.

2

Run

Return to the Terminal. Make sure the previous bat world is closed, then run:

```
python3 WorldOfBats.py
```

CHANGE IT If you want a different look, press **Q**, edit the three values, save, and run again.

How to read an eight-bit appearance code

1111	0100	11110100
SIZE	STYLE	COMPLETE CODE

4. Teach your bat to hunt

Edit only the indented lines under **while True:**. These instructions repeat for as long as your bat is flying.

METHOD	WHAT IT DOES
`setSpeed(number)`	Sets speed from 0.0 (slow) to 1.0 (fast).
`turnLeft()`	Turns a tiny amount to the left.
`turnRight()`	Turns a tiny amount to the right.
`ping()`	Looks straight ahead for the nearest insect using echolocation.

What ping() gives you

```
result = self.ping()
result["distance"] # 0 if nothing is seen; otherwise 1 to 127
result["size"] # 0 if nothing is seen; otherwise 2, 3 or 4
```

A starting strategy

```
while True:
self.setSpeed(0.8)
result = self.ping()
if result["distance"] == 0:
self.turnLeft()
```

ENERGY Flying uses energy. Faster flight uses energy more quickly. Eating larger insects restores more energy.

EXPERIMENT Try changing one thing at a time. Does speed help? How often should the bat turn? Can distance or insect size improve your decision?

5. Challenge another bat

1. **Find another team**

   Ask a team that has finished its bat if they would like to compete.

2. **Copy one whole bat folder**

   Copy the team's renamed folder into your own **BatCave**. Do not copy only MyBat.py.

3. **Restart the world**

   Close any running bat world with **Q**, then run **python3 WorldOfBats.py** again.

4. **Show scores**

   Press **Tab** while the world is running. After you press **Q**, final scores also appear in the Terminal.

THE GRAND BAT-OFF

How many different bats can you run in one world? Which hunting idea works best - and why?

Quick fixes

The Terminal says the folder does not exist	Check spelling and capital letters. Type **ls** to list files in the current folder.
The bat world does not open	Run **./run-on-linux.sh** again, then show the full Terminal message to a demonstrator.
My code gives an error	Check brackets, quotation marks and indentation. Compare with the starting code or ask a demonstrator.
My changes do not appear	Press **Ctrl+S**, close the old bat world with **Q**, then restart it.